



Red Pitaya Hardware Specifications V1.0.1

Product name: **Red Pitaya, model V1.0**

- RF inputs
 - Number of channels: 2
 - Bandwidth: 50 MHz (3 dB)
 - Sample rate: 125 Msps
 - ADC resolution 14 bits
 - Input coupling: DC
 - Input noise level: < -119 dBm /Hz (D)
 - Input impedance: $1\text{ M}\Omega // 10\text{ pF}$ (A,B)
 - Full scale voltage: 2Vpp, (46 Vpp for low-gain jumper setting) (T,V)
 - DC offset error: $<5\%$ FS (G)
 - gain error: $<3\%$ (at high gain jumper setting), $<10\%$ (at low gain jumper setting) (G)
 - Absolute maximum input voltage rating: 30 V (S) (1500 V ESD)
 - Overload protection: protection diodes (under the input voltage rating conditions)
 - Input channel isolation: typical performance 65 dB @ 10 kHz, 50 dB @ 100 kHz, 55 dB @ 1 M, 55 dB @ 10 MHz, 52 dB @ 20 MHz, 48 dB @ 30 MHz, 44 dB @ 40 MHz, 40 dB @ 50 MHz. (C)
 - Harmonics
 - at -3 dBFS: typical performance <-45 dBc (E)
 - at -20 dBFS: typical performance <-60 dBc (E)
 - Main spurious frequency components: Typically <-65 dBFS (at 2.3 MHz) (F)
 - Connector type: SMA (U)
 - Frequency response is adjusted by digital compensation
- RF outputs
 - Number of channels: 2
 - Bandwidth: 50 MHz (3 dB) (K)
 - Sample rate: 125 Msps
 - DAC resolution: 14 bits

- Output coupling: DC
- Load impedance: 50 Ω (J)
- Output slew rate limit: 200 V/us
- Connector type: SMA (U)
- DC offset error: < 5% FS (G)
- Gain error: < 5% (G)
- Full scale power: > 9 dBm (L)
- Harmonics: typical performance: (at -8 dBm)
 - -51 dBc @ 1 MHz
 - -49 dBc @ 10 MHz
 - -48 dBc @ 20 MHz
 - -53 dBc @ 45 MHz
- Auxiliary analog input channels:
 - Number of channels: 4
 - Nominal sampling rate: 100 ksps (H)
 - ADC resolution 12 bits
 - Connector: dedicated pins on IDC connector E2 (pins 13,14,15,16)
 - Input voltage range: 0 to +3.5 V
 - Input coupling: DC
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- Auxiliary analog output channels
 - Number of channels: 4
 - Output type: Low pass filtered PWM (I)
 - PWM time resolution: 4ns (1/250 MHz)
 - Connector: dedicated pins on IDC connector E2 (pins 17,18,19,20)
 - Output voltage range: 0 to +1.8 V
 - Output coupling: DC
- General purpose digital input/output channels: (N)
 - Number of digital input/output pins: 16
 - Voltage level: 3.3 V
 - Direction: configurable
 - Location: IDC connector E1 (pins 3-24)
- Extension:
 - Connector: 2 x 26 pins IDC (M)
 - Power supply:
 - Available voltages: +5V, +3.3V, -3.3V
 - Current limitations: 500 mA for +5V and +3.3V (to be shared between extension module and USB devices), 50 mA for -3.3V supply.
- Serial connectivity:
 - Connector type: 2x “SATA type” connector
 - Two differential pairs per connector enabling up to 500 Mbps
- CPU / FPGA: Xilinx Zynq 7010 SoC

- 17,600 LUTs
 - 28k Logic Cells
 - 80 DSP slices
 - Dual ARM® Cortex™-A9
- Memory: 4 Gb DDR3 SDRAM (512 MB)
- SD card:
 - Card type: micro SD
 - Supported standard: SD/SDIO 2.0/MMC3.31
 - File System: FAT32 (255 heads, 63 sectors)
 - Maximum card size: 32 GB (tested with: Apacer microSD HC 4GB Class 4 and others)
- WiFi:
 - WiFi USB dongles: tested with Edimax EW-7811Un
- Fan: Optional fan power supply (5V with 39 Ω source resistance)
- Power supply: (O)
 - Voltage: 5 V
 - Current: 2 A
 - Connector type: micro USB
 - Compliant with relevant regulation standards
 - Recommended power supply: HNP10I-microUSB
- Dimensions: 107 mm x 60 mm x 21 mm (W)
- Power consumption: typically < 0.9 A @ 5V (P)
- Environmental conditions:
 - Temperature:
 - Minimum ambient operating temperature: 5 °C
 - Maximum ambient operating temperature: 30 °C (P)
 - Storage temperature range from -50 °C to 150 °C
 - Laboratory ambient temperature range: 22 °C $\pm$ 2 °C (R)
 - Exposed surface temperature: < 70 °C (Q)
 - Maximum Zynq SoC IC operation temperature: 85 °C
 - Humidity: 15 % - 85 % (non condensing)

Notes:

- A. Input capacitance depends on jumper settings and may vary.
- B. A 50 Ω termination can be connected through an SMA tee in parallel to the input for measurements in a 50 Ω system.
- C. Crosstalk measured with high gain jumper setting on both channels. The SMA connectors not involved in the measurement are terminated.

- D. Measurement referred to high gain jumper setting, with limited environmental noise, inputs and outputs terminated, output signals disabled, PCB grounded through SMA ground. The specified noise floor measurement is calculated from the standard deviation of 16k contiguous samples at full rate. (Typically full bandwidth $\text{std}(V_n) < 2 \text{ mV}$). Noise floor specification does not treat separately spurious spectral components and represents time domain noise average referred to a 1 Hz bandwidth. In presence of spurious components the actual noise floor would result lower.
- E. Measurement referred at high gain jumper setting, inputs matched and outputs terminated, outputs signal disabled, PCB grounded through SMA ground.
- F. Measurement referred to high gain jumper setting, inputs and outputs terminated, outputs signal disabled, PCB grounded through SMA ground. Other less significant spurious components may be introduced by the RF front end.
- G. Further corrections can be applied through more precise gain and DC offset calibration.
- H. Default software enables sampling at CPU dependent speed. The acquisition of sequence at 100 kps rate requires the implementation of additional FPGA processing.
- I. First order low pass filter implementation. Additional filtering can be externally applied according to application requirements.
- J. The output channels are designed to drive 50 Ω loads. Terminate outputs when channels are not used. Connect parallel 50 Ω load (SMA tee junction) in high impedance load applications.
- K. Measured at -10 dBm output power level
- L. Typical power level with 1 MHz sine is 9.5 dBm. Output power is subject to slew rate limitations.
- M. Detailed scheme available within documentation (Red_Pitaya_Schematics_v1.0.1.pdf)
- N. To avoid speed limitations on digital General Purpose Input / Output pins are directly connected to FPGA. FPGA decoupling and pin protection is to be addressed within extension module designs. User is responsible for pin handling.
- O. The use of not approved power supply may deteriorate performance or damage the product.
- P. Heatsink must be installed and board must be operated on a flat surface without airflow obstructions. Operation at higher ambient temperatures, lower pressure conditions or within enclosures to be addressed by means of adequate ventilation. The operation of the product is automatically disabled at increased temperatures.
- Q. Some parts may become hot during and after operation. Do not touch them.
- R. Measurement performance is specified within this range.
- S. Valid for low frequency signals. For input signals that contain frequency components beyond 1 kHz, the full scale value defines the maximum admissible input voltage.
- T. Jumper settings are limited to the positions described in the user manual. Any other configuration or use of different jumper type may damage the product.
- U. SMA connectors on the cables connected to Red Pitaya must correspond to the standard MIL-C-39012. It's Important that central pin is of suitable length, otherwise the SMA connector installed in Red Pitaya will mechanically damage the SMA connector. Central pin of the SMA connector on Red Pitaya will loose contact to the board and the

board will not be possible to repair due to the mechanical damage (separation of the pad from the board).

- V. Jumpers are not symmetrical, they have latches. Always install jumpers with the latch on its outer side in order to avoid problems with hard to remove jumpers.
- W. Dimensions are rounded to the nearest millimeter. For exact dimensions, please see the Technical drawings and product model. (Red_Pitaya_Dimensions_v1.0.1.pdf)

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